

Electrochemistry and Corrosion

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13/2/25
Electrochemistry is a branch of Physical chemistry that deals with transformation of chemical energy into electrical energy and vice versa. An electrochemical process consists of oxidation and reduction reactions occurring at electrodes and is ~~is~~ accompanied by direct transfer of electrons through the phase-interface.

Difference Between Galvanic cells and Electrolytic cells

Galvanic cells	Electrochemical cells	Electrolytic cells
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- | | |
|--|---|
| 1. - A device used to convert chemical Energy into Electrical Energy | - A device used to convert Electrical Energy to Chemical Energy |
| - Negative electrode - Anode | Negative electrode - Cathode |
| - Positive electrode - Cathode | Positive electrode - Anode |
| - Electrons moves from anode to cathode | Electrons moves from Cathode to anode |

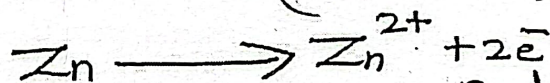
Cell reaction

- | | |
|--|--|
| - Ions discharged only on cathode | - Ions discharged on the both electrodes |
| - Concentration of anodic half cell rises while that of cathodic half cell decreases | - Concentration of Electrolyte decrease |

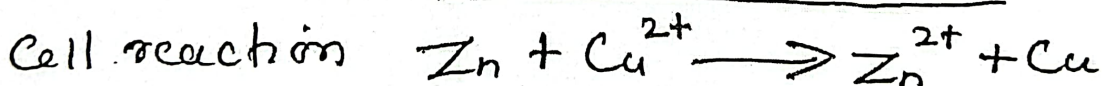
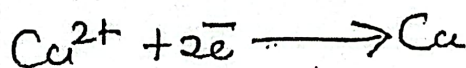
Daniel cell (2)

Daniel cell is an example of Galvanic cell. It consists of Zinc electrode dipping in $ZnSO_4$ solution (where oxidation takes place) and a copper electrode, dipping in $CuSO_4$ solution (where reduction takes place). Each electrode may be regarded as a half cell. The two solutions are separated by salt bridge.

The electrode reactions in Daniel cell are
At anode (-ve) electrode (oxidation)



At cathode +ve electrode (reduction)



The emf of cell is 1.1 volt

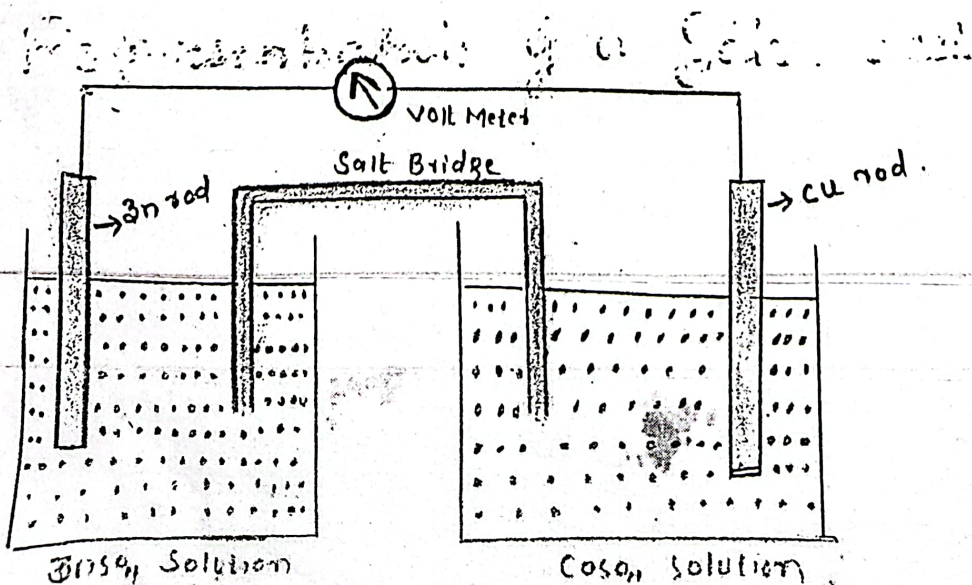


Fig. Daniel cell with Salt Bridge

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The electrode at which oxidation occurs is anode and at which reduction occurs is cathode. Due to oxidation anode acquires negative charge and at cathode due to reduction it acquires a positive charge.

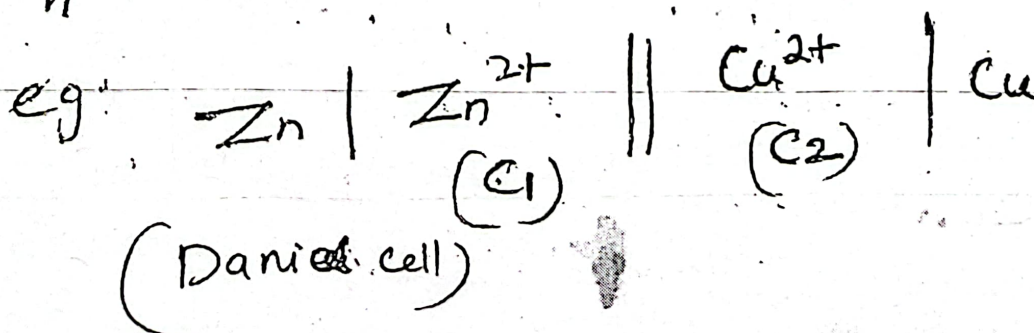
Functions of Salt bridge are

1. It completes the Electrochemical circuit with electrons flowing from one electrode to other through external wires and flow of ions between two compartments through the salt bridge.
2. It maintains electrical neutrality in the cell.

Representation of Galvanic cell

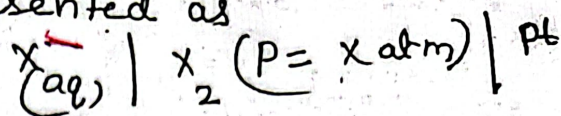
Electrode (anode)		ions in Solution		ions in Solution		Electrode (Cathode)
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|| - Vertical lines - represents the salt bridge.

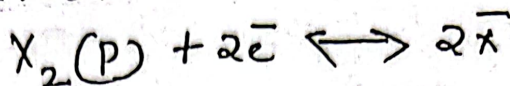


Q (4)

The electrode is represented as



The electrode rxn is



The potential of this electrode is

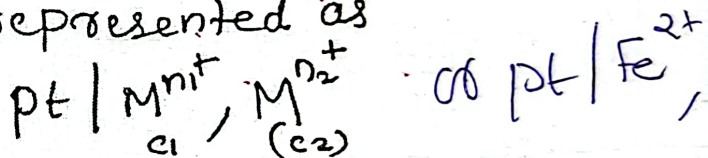
$$E = E^0_{\bar{X}|X_2|\text{Pt}} - \frac{2 \cdot 303RT}{nF} \log \frac{[\bar{X}]^2}{P_{X_2}}$$

eg: SHE, Chlorine Electrode.

Redox Electrode:-

It contains an inert electrode like platinum dipped in a solution containing ions of the same substance in two oxidation states (Fe^{3+} and Fe^{2+}).

The electrode can be represented as



The electrode rxn $M^{n2+} + ne^- \rightleftharpoons M^{n1+}$

The potential of Electrode is

$$E = E^0_{M^{n1+}, M^{n2+}} - \frac{2 \cdot 303RT}{nF} \log \frac{[M^{n1+}]}{[M^{n2+}]}$$

Reference Electrodes:-

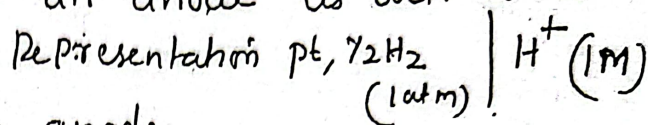
A reference electrode is one whose potential is known and hence can be used to compare the potential of other electrodes. Two Types of reference Electrodes.

(1) Primary Reference Electrodes (SHE)
Standard Hydrogen Electrode

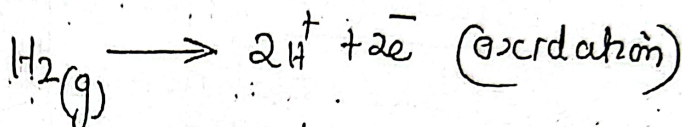
⑤

metal coated with platinum black kept immersed in a 1M solution of H^+ ions. (1M HCl). The hydrogen gas at a pressure of 1 atm is bubbled over the electrode at 298 K. The potential of SHE is taken as zero.

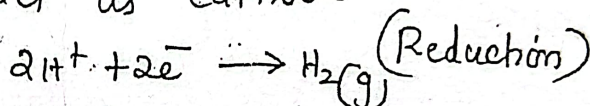
It can act as an anode as well as a cathode.



When it act as anode



when it act as cathode

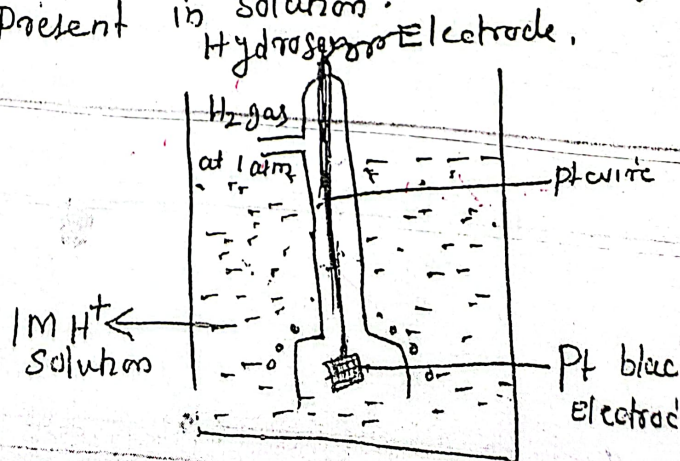


Merits of SHE

- Act as a primary reference electrode
- used to determine the pH of a solution.

Drawbacks of SHE (Demerits)

1. It is very difficult to maintain unit concentration of H^+ ion in solution
2. It is very difficult to maintain hydrogen gas at 1 atm. pressure uniformly.
3. This electrode can not be used if some oxidising and reducible ions present in solution.

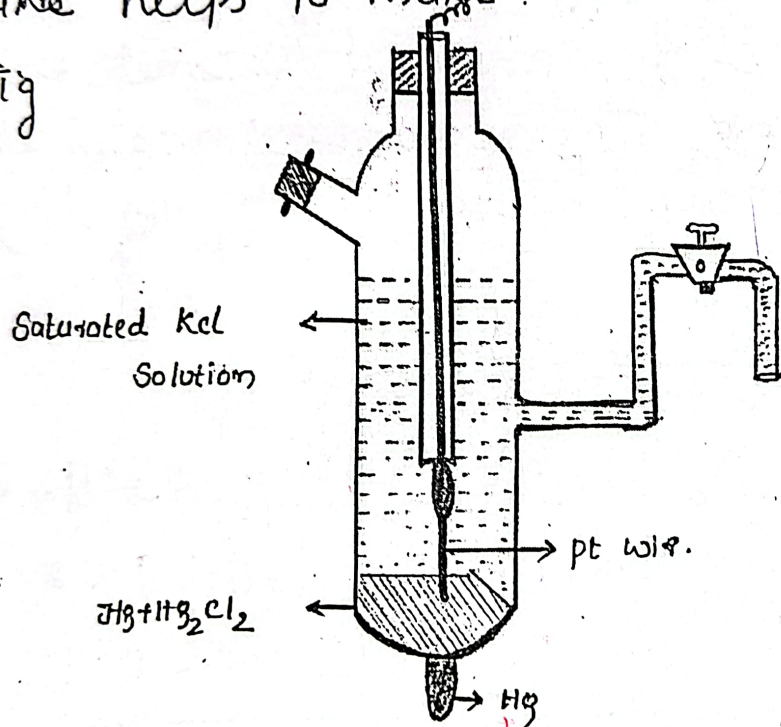


Secondary Reference Electrode

Saturated Calomel Electrode

It consists of a glass tube at the bottom of which is kept mercury, a paste of mercurous chloride (Calomel) and then filled with saturated solution of KCl. A platinum wire clipped into the glass tube ~~make~~ helps to make electrical contact.

Fig



Saturated
0.242

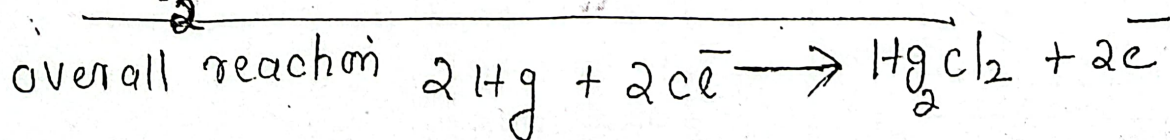
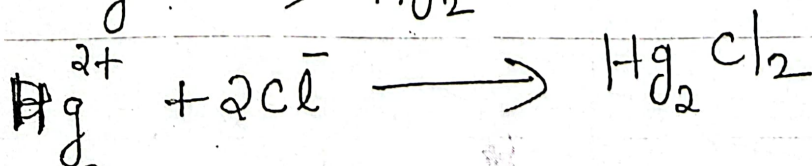
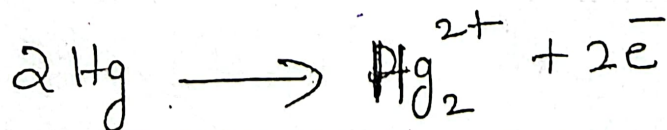
0.1M

Electrode potential
0.3335.

1M

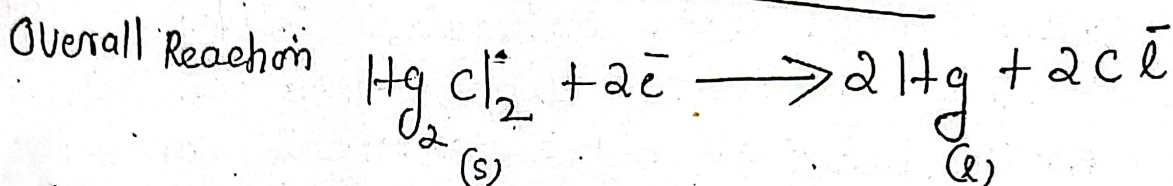
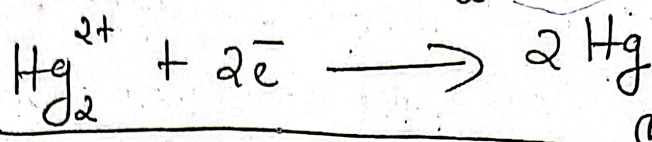
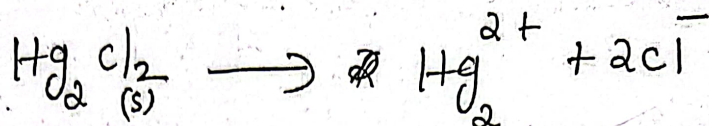
0.2810.

The following reactions take place when this electrode act as anode (oxidation occurs)



Standard electrode potential depend up on the conc. both used.

The following reaction take place when the electrode act as Cathode (Reduction occur)



The potential of saturated Calomel electrode is 0.2422 V. The saturated calomel electrode is represented as $\text{Hg}, \text{Hg}_2\text{Cl}_2; \text{KCl (saturated)}$

Merits

1. It is simple to construct.
2. Used to determine the pH of a solution.

Determination of E° (potential) using Saturated Calomel Electrode

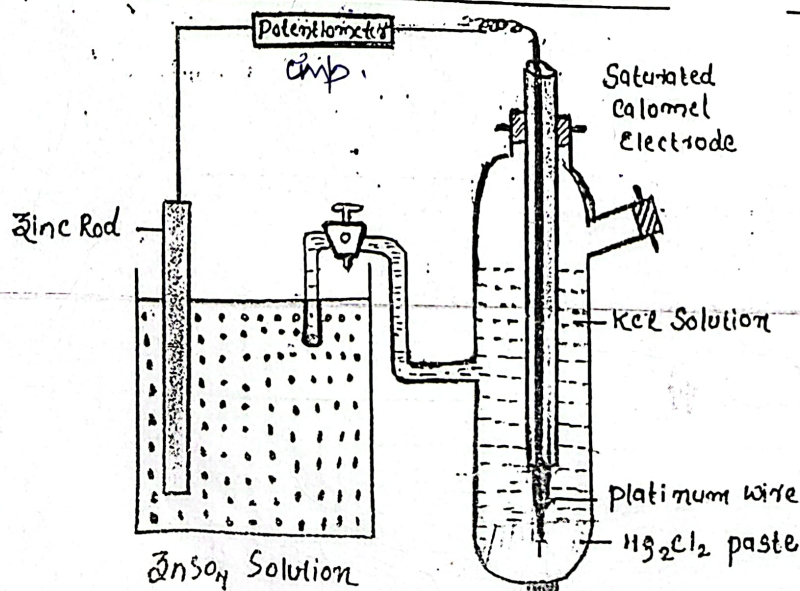


Fig: Measurement of Electrode potential.

(8)
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In order to determine the unknown potential of an electrode, the calomel electrode is coupled with that electrode and the emf of the resulting cell is determined using a potentiometer.

$$E_{\text{cell}}^{\circ} = E_{\text{calomel}}^{\circ} - E_{\text{Zn}}^{\circ}$$

E_{cell}° measured is 1V

$$\therefore 1V = E_{\text{calomel}}^{\circ} - E_{\text{Zn electrode}}^{\circ}$$

$$= 0.2422 - E_{\text{Zn electrode}}^{\circ}$$

$$E_{\text{Zn electrode}}^{\circ} = 0.2422 - 1 = \underline{\underline{-0.76V}}$$

Glass Electrode

The Glass electrode assembly consists of a thin glass bulb filled with $0.1M$ HCl into which a silver wire coated with silver chloride is immersed. Here Ag/AgCl acts as the internal reference electrode.

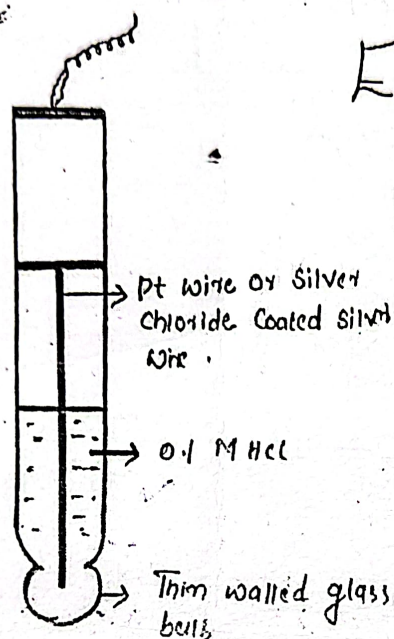
It is represented as $\text{Ag}/\text{AgCl}_{(s)} / 0.1M \text{HCl} / \text{glass}$

It works on the principle that when a thin glass membrane separates two solutions of different pH values, a potential develops across the glass membrane whose

magnitude depends upon the difference in pH of the two solutions

According to Nernst Equation

Electrode potential of Glass electrode



$$E_G = E_G^0 + 0.0591 \log [H^+]$$

$$E_G = E_G^0 - 0.0591 \text{ pH}$$

Glass Electrode

Merits of Glass Electrode

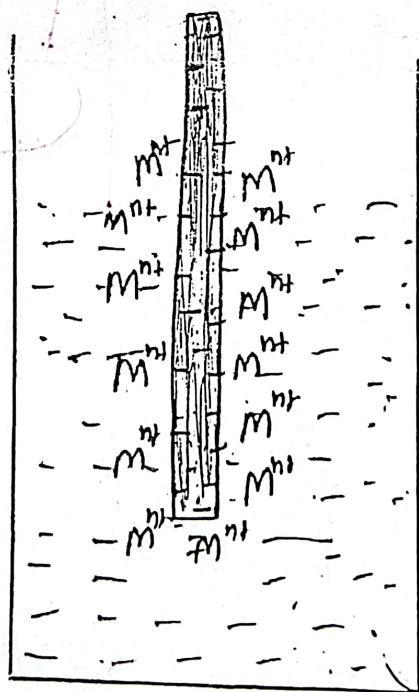
- It is simple to operate
- used to determine the pH of both oxidising and reducing solutions
- It is not easily poisoned.

Disadvantages

- special electronic potentiometers are used to determine the potential of an electrode

Single Electrode Potential

It is the measure of tendency of a metallic electrode to lose or gain electrons, when it is in contact with a 1M solution of its own ions at 25°C . The tendency of an electrode to lose electrons is a measure of its tendency to get oxidised and this tendency is called oxidation potential. Similarly, the tendency of an electrode to gain electrons is a measure of its tendency to get reduced and this tendency is known as reduction potential.



Helmholtz Double Layer

When a metal rod is in contact with a solution of its own ions, it may undergo either oxidation ($\text{M} \rightarrow \text{M}^{n+} + n\text{e}^{-}$) or reduction ($\text{M}^{n+} + n\text{e}^{-} \rightarrow \text{M}$) depending on its nature. Due to this either negative or positive charge

develops on the metal and the metal rod begins to attract Positive or negative charged free ions in solution. This leads to the formation of electrical double layer.

Due to this ~~electrical~~ formation of Helmholtz electrical double layer, a potential difference develops between the metal and the solution. This potential difference becomes a constant at Equilibrium and is called Standard electrode potential.

Electrochemical Series

If The electrodes ^{metals and nonmetals} are arranged in the increasing ~~or decreasing~~ ^{downwards} order of standard reduction potential or decreasing order of standard oxidation potential on hydrogen scale, it is known as electrochemical series.

Applications of Electrochemical Series

1. To predict relative ease of oxidation and reduction of different elements.

Greater is the reduction potential, more easily is the substance reduced, or it is a better oxidising agent. Thus we can say that Li is the best reducing agent and F is the weakest reducing agent.

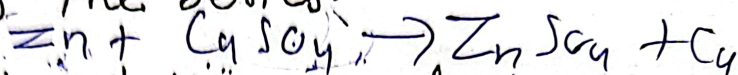
$$\Delta G^\circ = -nFE^\circ$$

(12)

$$\Delta G^\circ = -RT \ln K_{eq}$$

2 To predict a given metal will displace another from its salt solution

A metal higher in the series will displace the metal from its solution which is lower in the series.



3 To predict whether metal will evolve hydrogen on reaction with acid

All metals lying above hydrogen in the electrochemical series can react with acid to give hydrogen gas.

(4) To calculate Standard EMF of a cell.

$$\text{Standard EMF of a cell} = E^\circ_{\text{Cathode}} - E^\circ_{\text{Anode}}$$

5 To predict reducing power of metal

Reducing nature decreases from top to bottom in the electrochemical series

Reducing power depends on the tendency to ~~lose~~ lose e^- s.

6 To predict oxidising nature of non metal.

Oxidising nature depends on the tendency to accept e^- s. Thus oxidising nature increases from top to bottom in the Electro chemical Series.

Cells and Batteries

Secondary cells are those which can be recharged again and again by passing direct current through them. This recharging makes them suitable for reuse.

eg: Nickel Battery, Rechargeable Lithium-ion cell

Rechargeable Li-ion cell

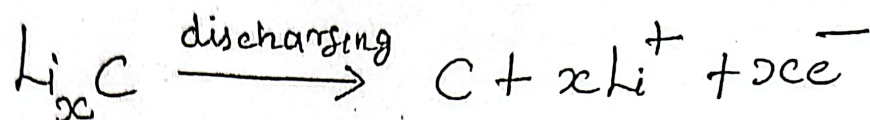
Lithium-ion cell consists of

- (a) Negative electrode - Lithiated Carbon intercalation compound
- (b) Positive electrode - Lithiated transition metal intercalation compound
- (c) Electrolyte - Li-salt ⁱⁿ Aprotic Organic Solvent

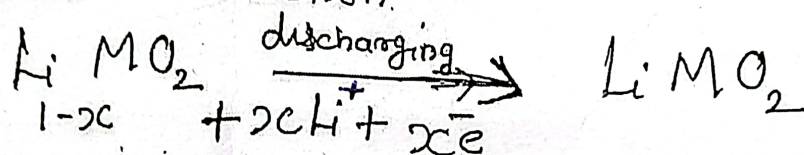
Discharging

During discharging, cell act as ~~an~~ a (electrochemical) Galvanic cell, the chemical energy is converted into electrical energy.

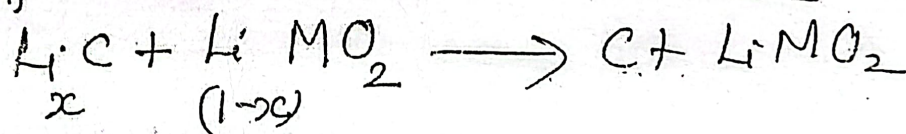
(-) Electrode Reaction



(+) Electrode Reaction



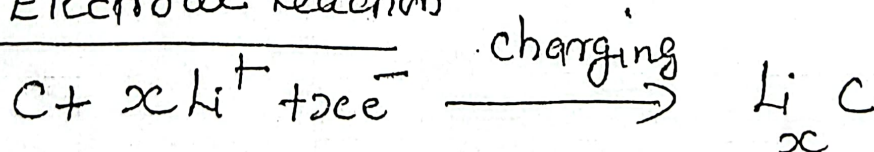
Net Reaction



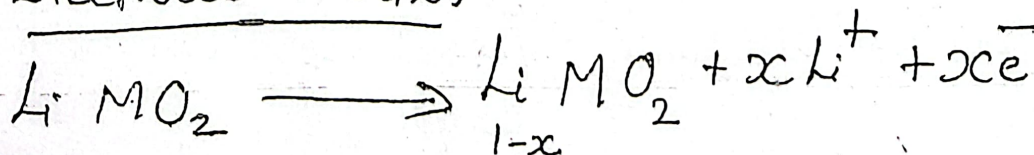
Charging

During charging cell act as an electrolytic cell, the electrical energy is converted into chemical energy

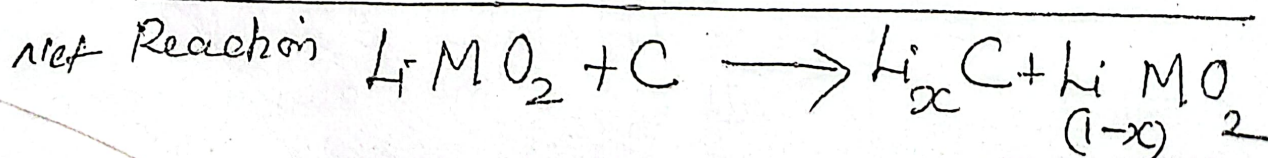
(-) Electrode Reaction



(+) Electrode Reaction



net Reaction



Lithium ions move back and forth between positive and negative electrode during charge and discharge. The electrochemical process is the uptake of lithium ions at the negative during charging and their release during discharge.

Overall Rxn

$$Li_xC + Li_{1-x}MO_2 \xrightleftharpoons[\text{Charging}]{\text{Discharging}} C + LiMO_2$$

Advantages

- (1) They are much lighter than other types of rechargeable batteries of same size.
- (2) They do not use Poisonous metals like lead, mercury or cadmium.

Applications

- (1) Lithium-ion cells are ideal for portable devices like note-book computers.
- (2) They are widely used in mobile phones and other electronic devices.
- (3) They find use in digital music devices like iPods, MP3 players.
- (4) They are used to power battery electric vehicles.
- (5) They are used in digital cameras, power tools etc.

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Conductivity

The substances which allows the passage of electric current through them are called conductors while those which do not allow flow of electric current are called insulators

- (1) Electronic conductors:- metals, semi-conductors
- (2) Electrolytic conductors - salt solution, acids, molten salt

Electrical Resistance

It is tendency to oppose the flow of electric current. Resistance of a conductor is proportional to its length (l) and inversely proportional to area (a) of ~~the~~ cross section of the conductor

$$R \propto \frac{l}{a}$$

$$R = \rho \frac{l}{a}$$

where ρ is called specific resistance or resistivity of the material

$$\rho = R \times \frac{a}{l} \quad \text{--- (1)}$$

Unit of ρ = ohm cm

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Conductance (C)

Conductance is the reciprocal of resistance

$$C = \frac{1}{R}$$

Unit of Conductance ohm^{-1} or mho

Specific Conductance (K), It is the
 reciprocal of specific resistance (ρ)
 or resistivity.

$$K = \frac{1}{\rho}$$

From (1)

$$K = \frac{1}{R} \times \frac{l}{a}$$

$$\text{Specific Conductance} = C \times \frac{l}{a}$$

(or Conductivity) = Conductance $\times$ Cell constant

Cell constant = $\frac{\text{Specific Conductance}}{\text{Conductance}}$

Unit of cell constant ($\frac{l}{a}$) = cm^{-1} or m^{-1}

Unit of specific conductance

Definition of Specific Conductance $K = \frac{1}{R} \frac{l}{a} = \text{ohm}^{-1} \text{cm}^{-1}$

$$K = \frac{1}{R} \frac{l}{a} = C \times \frac{l}{a} \quad \text{when } l = 1 \text{ cm} \\ a = 1 \text{ cm}^2$$

$$k = \frac{C \times l}{a}$$

$$= C \times 1 = C$$

specific conductance = Conductance.

Thus specific conductance is conductance of a solution taken in a cell in which the electrodes are unit distance apart and have unit area of cross-section.